

## SUMA DE PRODUCTOS

$$S_p = \underbrace{a \cdot b}_{=1} + \underbrace{a \cdot \bar{b} \cdot c}_{=1} + \underbrace{a \cdot \bar{b} \cdot \bar{c}}_{=1} \leftarrow \text{No está en forma canónica}$$

$$S_p = a \cdot b \cdot 1 + a \cdot \bar{b} \cdot c + a \cdot \bar{b} \cdot \bar{c}$$

$$S_p = a \cdot b \cdot (c + \bar{c}) + a \cdot \bar{b} \cdot c + a \cdot \bar{b} \cdot \bar{c}$$

$$\rightarrow S_p = \underbrace{a \cdot b \cdot c}_{\substack{a=1 \\ b=1 \\ c=1 \\ m_7}} + \underbrace{a \cdot b \cdot \bar{c}}_{\substack{a=1 \\ b=1 \\ c=0 \\ m_6}} + \underbrace{a \cdot \bar{b} \cdot c}_{\substack{a=1 \\ b=0 \\ c=1 \\ m_5}} + \underbrace{a \cdot \bar{b} \cdot \bar{c}}_{\substack{a=1 \\ b=0 \\ c=0 \\ m_0}} \leftarrow \text{Si está en forma canónica}$$

$$S_p = m_7 + m_6 + m_5 + m_0$$

| DECIMALES | BINARIOS |
|-----------|----------|
| 0         | 0        |
| 1         | 1        |
| 2         | 10       |
| 3         | 11       |
| 4         | 100      |
| 5         | 101      |
| 6         | 110      |
| 7         | 111      |

Si la suma de productos está en la forma canónica entonces está en forma normal disyuntiva.

$$FND \quad f(a,b,c) = a \cdot b \cdot c + a \cdot b \cdot \bar{c} + a \cdot \bar{b} \cdot c + a \cdot \bar{b} \cdot \bar{c}$$

8.4)

|   | ABC | X                  |
|---|-----|--------------------|
| 0 | 000 | 0                  |
| 1 | 001 | 1 ← m <sub>1</sub> |
| 2 | 010 | 0                  |
| 3 | 011 | 0                  |
| 4 | 100 | 1 ← m <sub>4</sub> |
| 5 | 101 | 1 ← m <sub>5</sub> |
| 6 | 110 | 0                  |
| 7 | 111 | 1 ← m <sub>7</sub> |

$$S_p = m_1 + m_4 + m_5 + m_7$$

$$S_p = \underbrace{\bar{A} \bar{B} C}_{\substack{A=0 \\ B=0 \\ C=1}} + \underbrace{\bar{A} B \bar{C}}_{\substack{A=1 \\ B=0 \\ C=0}} + \underbrace{A \bar{B} C}_{\substack{A=1 \\ B=0 \\ C=1}} + \underbrace{A B C}_{\substack{A=1 \\ B=1 \\ C=1}} \leftarrow \text{forma canónica}$$

8.1. Desarrollar la tabla de verdad de cada una de las siguientes expresiones suma de productos

(a)  $\bar{A}B + ABC + \bar{A}\bar{C} + \bar{A}\bar{B}C$  (b)  $\bar{X} + Y\bar{Z} + \bar{X}Y\bar{Z}$

Postulado 4:  $a + \bar{a} = 1$

a)  $\bar{A} \cdot B + A \cdot B \cdot \bar{C} + \bar{A} \cdot \bar{C} + A \cdot \bar{B} \cdot C$

$$\bar{A} \cdot B \cdot 1 + A \cdot B \cdot \bar{C} + \bar{A} \cdot \bar{C} \cdot 1 + A \cdot \bar{B} \cdot C$$

$$\bar{A} \cdot B \cdot (C + \bar{C}) + A \cdot B \cdot \bar{C} + \bar{A} \cdot \bar{C} \cdot (B + \bar{B}) + A \cdot \bar{B} \cdot C$$

$$\bar{A} B C + \bar{A} B \bar{C} + A B \bar{C} + \bar{A} \bar{C} B + \bar{A} \bar{C} \bar{B} + A \bar{B} C$$

$$\bar{A} B C + \bar{A} B \bar{C} + A B \bar{C} + \bar{A} \bar{C} B + \bar{A} \bar{C} \bar{B} + A \bar{B} C$$

| A | B | C | f(A,B,C) |                |
|---|---|---|----------|----------------|
| 0 | 0 | 0 | 1        | m <sub>0</sub> |
| 0 | 0 | 1 | 0        |                |
| 0 | 1 | 0 | 1        | m <sub>2</sub> |
| 1 | 1 | 1 | 1        | m <sub>7</sub> |

$$\bar{A}BC + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{C}B + A\bar{C}B + A\bar{B}C$$

$$\bar{A}BC + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$f(A,B,C) = \underbrace{\bar{A}BC}_{\substack{A=0 \\ B=1 \\ C=1 \\ m_3}} + \underbrace{\bar{A}\bar{B}\bar{C}}_{\substack{A=0 \\ B=1 \\ C=0 \\ m_2}} + \underbrace{A\bar{B}\bar{C}}_{\substack{A=1 \\ B=1 \\ C=0 \\ m_6}} + \underbrace{\bar{A}\bar{B}\bar{C}}_{\substack{A=0 \\ B=0 \\ C=0 \\ m_0}} + \underbrace{A\bar{B}\bar{C}}_{\substack{A=1 \\ B=0 \\ C=1 \\ m_5}}$$

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

$$b) \bar{X} + Y\bar{Z} + X\bar{Y}Z$$

$$\bar{X}.1 + Y\bar{Z}.1 + X\bar{Y}Z$$

$$\bar{X}(Y+\bar{Y}) + Y\bar{Z}(X+\bar{X}) + X\bar{Y}Z$$

$$\bar{X}Y + \bar{X}\bar{Y} + Y\bar{Z}X + Y\bar{Z}\bar{X} + X\bar{Y}Z$$

$$\bar{X}Y.1 + \bar{X}\bar{Y}.1 + XY\bar{Z} + \bar{X}Y\bar{Z} + X\bar{Y}Z$$

$$\bar{X}Y(Z+\bar{Z}) + \bar{X}\bar{Y}(Z+\bar{Z}) + XY\bar{Z} + \bar{X}Y\bar{Z} + X\bar{Y}Z$$

$$\bar{X}YZ + \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z + \bar{X}\bar{Y}\bar{Z} + XY\bar{Z} + \bar{X}Y\bar{Z} + X\bar{Y}Z$$

$$\bar{X}YZ + \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z + \bar{X}\bar{Y}\bar{Z} + XY\bar{Z} + X\bar{Y}Z$$

$$\underbrace{\bar{X}YZ}_{\substack{X=0 \\ Y=1 \\ Z=1 \\ m_3}} + \underbrace{\bar{X}Y\bar{Z}}_{\substack{X=0 \\ Y=1 \\ Z=0 \\ m_2}} + \underbrace{\bar{X}\bar{Y}Z}_{\substack{X=0 \\ Y=0 \\ Z=1 \\ m_1}} + \underbrace{\bar{X}\bar{Y}\bar{Z}}_{\substack{X=0 \\ Y=0 \\ Z=0 \\ m_0}} + \underbrace{XY\bar{Z}}_{\substack{X=1 \\ Y=1 \\ Z=0 \\ m_6}} + \underbrace{X\bar{Y}Z}_{\substack{X=1 \\ Y=0 \\ Z=1 \\ m_5}}$$

$$\bar{X}.1 = \bar{X}(Y+\bar{Y}) = \bar{X}Y + \bar{X}\bar{Y}$$

$$\bar{X}Y.1 + \bar{X}\bar{Y}.1$$

$$\bar{X}Y(Z+\bar{Z}) + \bar{X}\bar{Y}(Z+\bar{Z})$$

$$\bar{X}YZ + \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z + \bar{X}\bar{Y}\bar{Z}$$

| X | Y | Z | f(X,Y,Z) |
|---|---|---|----------|
|   |   |   |          |
|   |   |   |          |
|   |   |   |          |
|   |   |   |          |
|   |   |   |          |
|   |   |   |          |

## PRODUCTO DE SUMAS

$$P_S = (\underbrace{a+\bar{c}}_{=0}) \cdot (\underbrace{a+\bar{b}+c}_{=0}) \cdot (\underbrace{\bar{a}+\bar{b}+\bar{c}}_{=0})$$

No está en forma canónica

Para que el producto de sumas esté en forma canónica, tenemos que tener productos de Maxi términos

$$P_S = (a+\bar{c}+0) \cdot (a+\bar{b}+c) \cdot (\bar{a}+\bar{b}+\bar{c})$$

$$P_S = (a+\bar{c}+(\bar{b} \cdot \bar{b})) \cdot (a+\bar{b}+c) \cdot (\bar{a}+\bar{b}+\bar{c})$$

$$P_S = (a+\bar{c}+\bar{b}) \cdot (a+\bar{c}+\bar{b}) \cdot (a+\bar{b}+c) \cdot (\bar{a}+\bar{b}+\bar{c})$$

$$P_S = (a+\bar{c}+\bar{b}) \cdot (a+\bar{c}+\bar{b}) \cdot (a+\bar{b}+c) \cdot (\bar{a}+\bar{b}+\bar{c})$$

Está en forma canónica

### MAXI TÉRMINOS

$$M_i = \sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$$

$$M_i = \sum_{i=1}^5 a_i = \underbrace{a_1 + a_2 + a_3 + a_4 + a_5}_{a + b + c + d + e}$$

$$P_5 = (a+b+\bar{c}).(a+\bar{b}+\bar{c}).(a+\bar{b}+c).(\bar{a}+\bar{b}+\bar{c})$$

FORMA NORMAL CONJUNTIVA: producto de sumas está en forma canónica

$$F(a,b,c) = (a+b+\bar{c}).(a+\bar{b}+\bar{c}).(a+\bar{b}+c).(\bar{a}+\bar{b}+\bar{c})$$

Ej:  $a\bar{b}+c$

| a | b | c | $\bar{b}$ | $a\bar{b}$ | $a\bar{b}+c$ |
|---|---|---|-----------|------------|--------------|
| 0 | 0 | 0 | 1         | 0          | 0            |
| 0 | 0 | 1 | 1         | 0          | 1            |
| 0 | 1 | 0 | 0         | 0          | 0            |
| 0 | 1 | 1 | 0         | 0          | 1            |
| 1 | 0 | 0 | 1         | 1          | 1            |
| 1 | 0 | 1 | 1         | 1          | 1            |
| 1 | 1 | 0 | 0         | 0          | 0            |
| 1 | 1 | 1 | 0         | 0          | 1            |

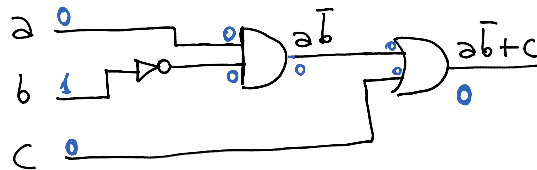
Nos fijamos en la tabla en los 0s, de ahí salen los Maxitérminos.

Los 0 no se complementan  
Los 1 si se complementan

$$P_5 = M_0 \cdot M_2 \cdot M_6$$

$$P_5 = \underbrace{(a+b+\bar{c})}_{\substack{a=0 \\ b=0 \\ c=0}} \cdot \underbrace{(a+\bar{b}+\bar{c})}_{\substack{a=0 \\ b=1 \\ c=0}} \cdot \underbrace{(\bar{a}+\bar{b}+\bar{c})}_{\substack{a=1 \\ b=1 \\ c=0}}$$

$$F(a,b,c) = (a+b+\bar{c}).(a+\bar{b}+\bar{c}).(\bar{a}+\bar{b}+\bar{c})$$



- Producto de sumas, para que estén en forma canónica den estar como productos de Maxitérminos } FNC
- Cada Maxitérmino da 0
- Altera las variables con 0 no se complementan y las variables con 1 si se complementan

En los Maxitérminos se suman las variables

8.4

| ABC | X                  |
|-----|--------------------|
| 000 | 0 ← M <sub>0</sub> |
| 001 | 1                  |
| 010 | 0 ← M <sub>2</sub> |
| 011 | 0 ← M <sub>3</sub> |
| 100 | 1                  |
| 101 | 1                  |
| 110 | 0 ← M <sub>6</sub> |
| 111 | 1                  |

$$P_5 = M_0 \cdot M_2 \cdot M_3 \cdot M_6$$

$$P_5 = \underbrace{(A+B+C)}_{\substack{A=0 \\ B=0 \\ C=0}} \cdot \underbrace{(A+\bar{B}+C)}_{\substack{A=0 \\ B=1 \\ C=0}} \cdot \underbrace{(A+\bar{B}+\bar{C})}_{\substack{A=0 \\ B=1 \\ C=1}} \cdot \underbrace{(\bar{A}+\bar{B}+\bar{C})}_{\substack{A=1 \\ B=1 \\ C=0}}$$